

1. Hardware (IoT)

Component	Purpose	Approx. Price (₹)
ESP32 Dev Board	Main IoT controller	500–800
ESP32-CAM	AI camera for drowsiness/helmet monitoring	700–1,000
MPU6050	Accident & tilt detection	120–200
MAX30102	Heart Rate & SpO ₂ sensor	250–450
MQ-3 Alcohol Sensor	Alcohol detection	180–350
NEO-6M GPS Module	Live location	600–900
SIM800L GSM Module	SMS & emergency calls	600–900
HC-05 Bluetooth	Phone communication (optional)	250–350
Vibration Motor	Haptic alerts	50–100
Buzzer	Audio alerts	20–50

Li-ion 18650 Battery (2×)	Helmet power	300–500
TP4056 Charging Module	Battery charging	50–100
Battery Holder & Switch	Power management	80–150
Jumper Wires & PCB	Connections	200–400
Helmet	Base helmet	1,200–2,500

Estimated Hardware Cost: ₹5,500 – ₹9,000

2. AI Stack

Technology	Purpose	Cost
Python	AI development	Free
OpenCV	Image processing	Free
MediaPipe	Face & eye tracking	Free
TensorFlow Lite	AI inference	Free

YOLOv11	Object detection	Free Source)	(Open
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3. IoT Stack

Technology	Purpose	Cost
Arduino IDE	ESP32 programming	Free
ESP-IDF (Optional)	Advanced development	ESP32 Free
MQTT	IoT communication	Free
Wi-Fi	Internet connectivity	Free

4. Backend

Since you already know the **MERN Stack**, this is a great fit.

Technology	Purpose	Cost
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Node.js	Backend server	Free
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Express.js	REST API	Free
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Socket.IO	Real-time communication	Free
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5. Database

Technology	Purpose	Cost
MongoDB Atlas	Store users, rides, alerts	Free Tier
Firebase (Optional)	Firestore Real-time sync	Free Tier

6. Frontend

Technology	Purpose	Cost
React.js	Web dashboard	Free
Tailwind CSS	UI styling	Free

Chart.js	Graphs & analytics	Free
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Leaflet.js or Google Maps API	Live location	Free (with usage limits)
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7. Mobile App

Since you're familiar with JavaScript:

Technology	Purpose	Cost
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React Native	Android app	Free
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Alternative: Flutter (also free) if you're comfortable with Dart.

8. Cloud & Deployment

Service	Purpose	Cost
Render	Backend hosting	Free Tier
Vercel	Frontend hosting	Free Tier
MongoDB Atlas	Database	Free Tier

Firebase	Notifications/Auth	Free Tier
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9. APIs

API	Purpose	Cost
Google Maps API	Maps & navigation	Free tier available
OpenWeather API	Weather alerts	Free tier
Twilio (Optional)	SMS	Paid after free credits

Total Estimated Cost

Version	Estimated Cost
Basic Prototype	₹5,500 – ₹7,000
Recommended Final-Year Project	₹8,000 – ₹10,000

Advanced Version (extra sensors, better battery, enclosure) ₹10,000 –
₹15,000

Recommended Tech Stack (Best Choice)

Hardware

- ESP32
- ESP32-CAM
- MPU6050
- MAX30102
- MQ-3
- GPS NEO-6M
- SIM800L
- Vibration Motor
- Buzzer
- Li-ion Battery

AI

- Python
- OpenCV
- MediaPipe
- YOLOv11
- TensorFlow Lite

IoT

- Arduino IDE
- MQTT
- Wi-Fi

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

Frontend

- React.js
- Tailwind CSS
- Leaflet.js (or Google Maps)

Mobile App

- React Native
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Cost-Saving Tip

Don't buy everything at once. Start with the **core hardware**:

- ESP32
- ESP32-CAM
- MPU6050
- MQ-3
- GPS
- SIM800L

This will cost about **₹3,000–₹4,000** and is enough to build the core prototype. Later, add sensors like **MAX30102** and other advanced features as your project progresses. This staged approach also makes debugging much easier.